Sentiment Analysis of IMDB Movie Reviews

A Comparative Study of Machine Learning, Deep Learning, and Transformer Models

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Abstract

This report presents a comprehensive study on sentiment analysis applied to the IMDB Movie Reviews dataset, which contains 50,000 labeled reviews (49,582 after removing duplicates). The primary objective is to classify reviews as positive or negative using six models from three families: classical machine learning (Logistic Regression, Linear SVC, Naive Bayes), recurrent deep learning (Simple RNN and LSTM), and a pre-trained transformer (DistilBERT). Each model family is paired with a tailored preprocessing pipeline: aggressive for TF-IDF, light for sequence models, and minimal for the transformer. Linear SVC achieved the highest accuracy and weighted F1-score overall (91% and 0.91), while DistilBERT (90% accuracy, 0.90 weighted F1) achieved the highest recall on positive reviews (94%) at the cost of lower recall on negative reviews (87%). The LSTM matched Logistic Regression at 90% accuracy, outperforming the Simple RNN (83%) significantly. Differences of about one percentage point among the top four models are within the sampling noise of a single run on a 9,917-review test set.

1. Introduction

Sentiment analysis, a core task in Natural Language Processing (NLP), involves classifying text by its expressed opinion, typically positive or negative. Movie review platforms generate millions of opinionated texts daily, and automatically identifying their sentiment enables applications ranging from recommendation systems to brand monitoring.

The IMDB dataset is a standard benchmark in NLP, consisting of 25,000 positive and 25,000 negative reviews (24,884 positive and 24,698 negative after removing duplicates). This project explores how different model families from simple bag-of-words classifiers to state-of-the-art transformers compare on this task, and investigates the preprocessing strategies best suited to each.

1.1 Objectives

- Design and compare three preprocessing pipelines tailored to ML, DL, and Transformer models.
- Train and evaluate six models across three model families.
- Analyze trade-offs between model complexity, training cost, and performance.
- Identify the best model for production deployment on this task.

2. Dataset

The dataset used is the IMDB Dataset of 50K Movie Reviews, publicly available on Kaggle. It is one of the most widely cited benchmarks for binary sentiment classification.

Attribute	Details
Total Reviews	50,000
Positive Reviews	25,000 (50%); 24,884 after deduplication
Negative Reviews	25,000 (50%); 24,698 after deduplication
Duplicates Removed	418 exact duplicate reviews removed
Final Dataset Size	49,582 reviews after deduplication

Train / Test Split	80% / 20% (39,665 train | 9,917 test), stratified
Avg. Review Length	Mean 227 / median 171 words
95th Pct. Length	581 words (580 used as max sequence length for DL)

2.1 Key EDA Findings

- The dataset is nearly balanced (50.2% positive, 49.8% negative after deduplication), so accuracy is a fair metric.
- Review lengths vary widely: the median is 171 words, the 99th percentile is 893 words, and the longest review has 2,450 words.
- Positive and negative reviews have almost the same typical length (mean 229 vs 225 words; median 170 vs 171), although positive reviews are somewhat longer in the upper tail (95th percentile 597 vs 562 words).
- Top positive words: film, great, story, character, performance, love, best, beautiful.
- Top negative words: bad, worst, boring, waste, awful, terrible, poor, nothing.
- Bigram analysis confirms strong polarizing patterns: ‘highly recommend’, ‘must see’ (positive) vs ‘waste time’, ‘bad acting’ (negative).
- Top positive trigrams: ‘one best films’, ‘one best movies’, ‘well worth watching’; top negative trigrams: ‘worst movie ever’, ‘waste time money’, ‘bad acting script’.
- Vocabulary overlap analysis: 74,468 unique words in positive reviews and 71,829 in negative; 45,558 are shared across both classes, with 28,910 exclusive to positive and 26,271 exclusive to negative.
- Review length is only weakly related to sentiment: the share of positive reviews stays between about 49% and 55% for reviews up to 1,000 words, with no monotonic trend (the 1,000+ bucket reaches 69% but contains only 42 reviews).

2.2 Text Length & Vocabulary Analysis

Four length-related features were computed after stripping HTML and URLs: word count, character count, sentence count (punctuation-based), and average word length. Their distributions are summarized below.

- **Word Count:** Mean 227, Median 171, 75th pct 276, 90th pct 444, 95th pct 581, 99th pct 893 (maximum 2,450). The distribution is heavily right-skewed; half of all reviews fall between 125 and 276 words (interquartile range).
- **Character Count:** Mirrors the word count pattern with a median of 954 characters (mean 1,286).
- **Sentence Count:** Estimated via punctuation markers (. ! ?). The median review has 12 sentences (mean 14.1).
- **Average Word Length:** Approximately 4.6 characters per word (mean 4.64, interquartile range 4.41–4.84). This metric varies very little between reviews, so vocabulary sophistication (word length) is unlikely to be a strong differentiator.

2.3 Vocabulary Overlap

A vocabulary overlap analysis was performed on post-cleaned, stop-word-removed tokens (minimum 3 characters). The results quantify how much lexical overlap exists between the two sentiment classes.

- **Positive vocabulary:** 74,468 unique words (after stop-word removal, min length 3).
- **Negative vocabulary:** 71,829 unique words.
- **Shared words:** 45,558 words (about 45% of the 100,739-word combined vocabulary) appear in both classes. The large exclusive sets below probably consist largely of rare terms such as names and misspellings (not analyzed further), so word frequency and context, rather than exclusive vocabulary, carry most of the sentiment signal.
- **Exclusive to positive:** 28,910 words.
- **Exclusive to negative:** 26,271 words. The two exclusive vocabularies are of similar size, so there is no strong lexical asymmetry between the classes.

2.4 Review Length vs. Sentiment Heatmap

Reviews were bucketed into seven word-count ranges and the positive-class percentage was computed per bucket. The stacked bar chart shows no consistent trend: the proportion of positive reviews stays close to 50% in every bucket except the very small 1,000+ bucket.

- **<100 words:** 54% positive (6,452 reviews) — slightly positive-leaning.
- **100–500 words:** 49–50% positive (100–200: 49.1%, 200–300: 49.0%, 300–500: 49.6%); mid-length reviews are almost perfectly balanced.
- **500–1,000 words:** 54.6% positive (500–700) and 53.5% (700–1,000). The 1,000+ bucket is 69% positive but contains only 42 reviews, so it is not reliable.

Note: The weak length-sentiment relationship is a property of the raw data and does not impact the 80/20 stratified split, which preserves the class ratio (approximately 50/50) in both train and test sets.

3. Preprocessing

Three separate preprocessing pipelines were implemented, each optimized for the model family it feeds. A universal cleaning layer (HTML removal, URL removal, bracket stripping) is shared across all pipelines.

Pipeline	Lowercase	Stopwords Removed	Lemmatized	Notes
ML (TF-IDF)	Yes	Yes*	Yes (POS-aware)	*Negations preserved
DL Sequence	Yes	No	Yes (POS-aware)	Context words kept for RNN/LSTM
Transformer	No	No	No	WordPiece handles subwords

3.1 ML Preprocessing Pipeline

The ML pipeline applies the most aggressive transformations to maximize the effectiveness of TF-IDF sparse representations. Steps are applied in the following order:

- Remove HTML tags using BeautifulSoup.
- Remove square bracket content (e.g., [spoiler]).
- Remove URLs.

- Convert to lowercase.
- Expand contractions (e.g., don't → do not) using a curated 35-entry dictionary.
- Remove special characters and normalize whitespace.
- Tokenize using NLTK ToktokTokenizer.
- Remove stop words, preserving 21 negation words (not, never, can't, etc.) critical for sentiment polarity.
- POS-aware lemmatization using WordNetLemmatizer, mapping Penn Treebank POS tags to WordNet constants.

3.2 DL Sequence Pipeline (RNN / LSTM)

The DL sequence pipeline applies light preprocessing. Stop words are intentionally preserved because the embedding layer and recurrent units benefit from full contextual sequences. The pipeline omits stop-word removal but keeps POS-aware lemmatization.

3.3 Transformer Pipeline (DistilBERT)

The transformer pipeline is minimal; only HTML tags, URLs, and extra whitespace are removed. Casing, contractions, and stop words are preserved because DistilBERT's WordPiece tokenizer was pre-trained on raw text and handles subword segmentation internally. Aggressive preprocessing would discard information that the model was specifically trained to use.

4. Modeling

4.1 Machine Learning Models

Classical ML models were trained on TF-IDF vectorized text using unigrams and bigrams (ngram_range=(1,2)). Two vocabulary sizes were compared: 10,000 and 50,000 features. Results reported below use 50k features, which consistently outperformed the 10k configuration across all three models.

4.1.1 Logistic Regression

Logistic Regression with L2 regularization (max_iter=1000). A well-established baseline for text classification that models the log-odds of the positive class as a linear combination of TF-IDF features.

4.1.2 Linear SVC

Linear Support Vector Classifier finds the maximum-margin hyperplane in the TF-IDF feature space. Particularly effective for high-dimensional sparse text representations. Best-performing classical model.

4.1.3 Naive Bayes (Multinomial)

Multinomial Naive Bayes models word frequency distributions per class using Bayes' theorem. It is computationally the simplest model (training time was not benchmarked), but its conditional independence assumption limits performance on reviews with complex phrasing.

4.2 Deep Learning Models

Both DL models use a shared tokenization and padding pipeline. A Keras Tokenizer was fit on the training set with vocab_size=50,000 and oov_token='<OOV>'. Sequences were padded/truncated to max_length=580 (about the 95th percentile of review lengths: 581 words in the EDA, 597 on the lemmatized training set). An EarlyStopping callback (patience=3, restore_best_weights=True) was used to prevent overfitting; it monitored val_loss on the held-out test set (see Section 6.3).

4.2.1 Simple RNN

Architecture: Embedding(50000, 64, mask_zero=True) → SimpleRNN(64) → Dense(16, relu) → Dropout(0.1) → Dense(1, sigmoid). Trained for up to 15 epochs, batch_size=64, Adam(lr=0.001); early stopping ended training at epoch 10 and restored the weights from epoch 7. Simple RNNs suffer from vanishing gradients on long sequences, which explains the weaker performance compared to LSTM.

4.2.2 LSTM

Architecture: Embedding(50000, 64, mask_zero=True) → LSTM(64) → Dense(32, relu) → Dropout(0.3) → Dense(1, sigmoid). Trained for up to 10 epochs, batch_size=64, Adam(lr=0.0001); early stopping ended training at epoch 7 and restored the weights from epoch 4. LSTM gates (input, forget, output) address the vanishing gradient problem, enabling the model to capture long-range dependencies in reviews — the deciding factor in its superior performance over Simple RNN.

4.3 Transformer — DistilBERT Fine-Tuning

DistilBERT (distilbert-base-uncased) is a distilled version of BERT, retaining 97% of its language understanding while being 40% smaller and 60% faster. The model was fine-tuned end-to-end on the IMDB training set using the HuggingFace Trainer API.

Key hyperparameters:

- Max sequence length: 256 tokens (roughly 190 words). Longer reviews are truncated; since the 75th percentile is 276 words, more than a quarter of reviews exceed this length.
- Epochs: up to 8 (EarlyStopping on eval_loss, patience 2). Training actually stopped after 3 epochs (1,860 steps, about 14.5 minutes), which implies the best eval_loss was reached at epoch 1.
- Learning rate: 2e-5 (standard BERT fine-tuning range: 1e-5 to 5e-5).
- Batch size: 64 per device.
- Weight decay: 0.05.
- Warmup: the notebook sets warmup_steps=0.1, intended as a 10% warmup ratio (which would be warmup_ratio=0.1); see Section 6.3.
- Mixed precision (fp16) enabled when GPU is available.
- Dynamic padding via DataCollatorWithPadding for memory efficiency.

5. Results & Evaluation

All models were evaluated on the same held-out test set of 9,917 reviews (4,940 negative, 4,977 positive). The following table summarizes classification performance (precision, recall, and F1 are weighted averages over the two classes).

Model	Accuracy	Precision	Recall	F1-Score	Notes
Logistic Regression	90%	90%	90%	0.90	TF-IDF 50k
Linear SVC	**91%**	**91%**	**91%**	**0.91**	**TF-IDF 50k — highest accuracy and F1**
Naive Bayes	88%	88%	88%	0.88	TF-IDF 50k
Simple RNN	83%	83%	83%	0.83	Weakest model
LSTM	90%	90%	90%	0.90	Best DL sequence model
DistilBERT	90%	91%	90%	0.90	Highest positive recall (94%); negative recall 87%

5.1 Key Observations

- Linear SVC is the strongest model overall by accuracy (91%; 90.5% unrounded) and weighted F1 (0.91), and a TF-IDF plus linear model is lightweight, making it highly practical for production (training time was not formally benchmarked).
- LSTM significantly outperforms Simple RNN (90% vs 83%), demonstrating the importance of gating mechanisms for long-text sentiment.
- DistilBERT achieves the highest positive-class recall (94%; 319 positive reviews missed) but the lowest negative-class recall among the top four models (87%; 639 negative reviews classified as positive).
- Simple RNN is the only model that falls below 85%, making it unsuitable for this task without architectural improvements.
- Increasing TF-IDF features from 10k to 50k consistently improved ML model performance, indicating rich vocabulary coverage matters.
- The transformer did not outperform Linear SVC in accuracy (90.3% vs 90.5%, a difference of 19 reviews out of 9,917, well within sampling noise), but its recall advantage on positives, paid for with lower negative recall, may matter in asymmetric-cost scenarios.

5.2 Confusion Matrix Analysis

Analyzing the confusion matrices shows that Logistic Regression, Linear SVC, Naive Bayes, LSTM, and DistilBERT all make more false positives (negative reviews predicted positive) than false negatives, a common pattern in IMDB data due to the prevalence of mixed-sentiment language in critical reviews. DistilBERT shows this skew most strongly (639 false positives vs 319 false negatives), the other side of its 94% positive recall. Simple RNN is the exception: it makes more false negatives (1,051) than false positives (673).

6. Discussion

6.1 Model Selection Recommendations

The right model depends on the deployment context:

- Production (latency-sensitive): Linear SVC, 91% accuracy with a lightweight TF-IDF plus linear model (training and inference times were not formally benchmarked).
- Production (positive-recall-sensitive): DistilBERT has the best positive recall (94%) but lower negative recall (87%); its accuracy is on par with Linear SVC, so its extra cost is justified only when missing positive reviews is costlier than false alarms.
- Baseline / interpretability: Logistic Regression coefficients are directly interpretable as feature importances.
- Do not deploy Simple RNN: its 83% accuracy is significantly below alternatives with similar training complexity.

6.2 Effect of Preprocessing Choices

- Preserving negation words in the ML stopword filter is critical removing 'not' would flip the polarity of hundreds of phrases.
- POS-aware lemmatization provides more accurate root forms than stemming (e.g., 'better' → 'good' as adjective, not 'bett').
- Keeping stop words for DL sequence models preserves the grammatical structure that RNNs and LSTMs rely on for contextual encoding.
- The transformer preprocessing is deliberately minimal DistilBERT's pre-training on raw text means any additional cleaning risks removing information the model was trained to leverage.

6.3 Limitations

- Results are specific to the IMDB domain; sentiment models often require domain adaptation for other review types.
- The Simple RNN's weakness may be addressable with bidirectional architecture or attention mechanisms.
- DistilBERT was configured for up to 8 epochs, but early stopping ended training after 3 epochs (3 epochs is the standard recommendation for BERT-family models), so overfitting from a long run was avoided.
- No systematic hyperparameter search was performed (only the TF-IDF vocabulary size, 10k vs 50k, was compared).
- Model selection used the test set: EarlyStopping (RNN/LSTM) and load_best_model_at_end with EarlyStoppingCallback (DistilBERT) monitored the same held-out test set reported in Section 5, and the 10k vs 50k TF-IDF comparison was also made on it. Reported metrics may therefore be slightly optimistic; a separate validation split is recommended.
- Results come from a single run without confidence intervals. With 9,917 test reviews, the 95% interval on accuracy is roughly ±0.6 percentage points, so differences of about one point among the top models are not statistically meaningful.

- DistilBERT used max_length=256 tokens (truncating the longer reviews), and warmup was set with warmup_steps=0.1 instead of warmup_ratio=0.1, which may effectively disable warmup depending on the transformers version. Both may have limited its accuracy.

7. Conclusion

This project demonstrates that effective sentiment analysis on IMDB reviews can be achieved across a spectrum of model families, each with distinct trade-offs. Classical models with TF-IDF particularly Linear SVC remain highly competitive and are considerably cheaper to train than deep models. The LSTM substantially improves over Simple RNN, confirming the value of gated architectures for long-form text. DistilBERT matched, but did not exceed, the best classical model (90.3% vs 90.5% accuracy); its distinguishing feature is higher recall on positive reviews at the cost of lower recall on negative reviews. Its accuracy is probably limited by 256-token truncation and a short fine-tuning run, so it should not be read as the ceiling for this dataset.

The preprocessing pipeline design proved as important as model selection: tailoring cleaning, tokenization, and feature representation to each model family was essential to achieving competitive results. Future work could explore ensemble methods combining TF-IDF classifiers with transformer representations, data augmentation for harder examples, and cross-domain evaluation.

References

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